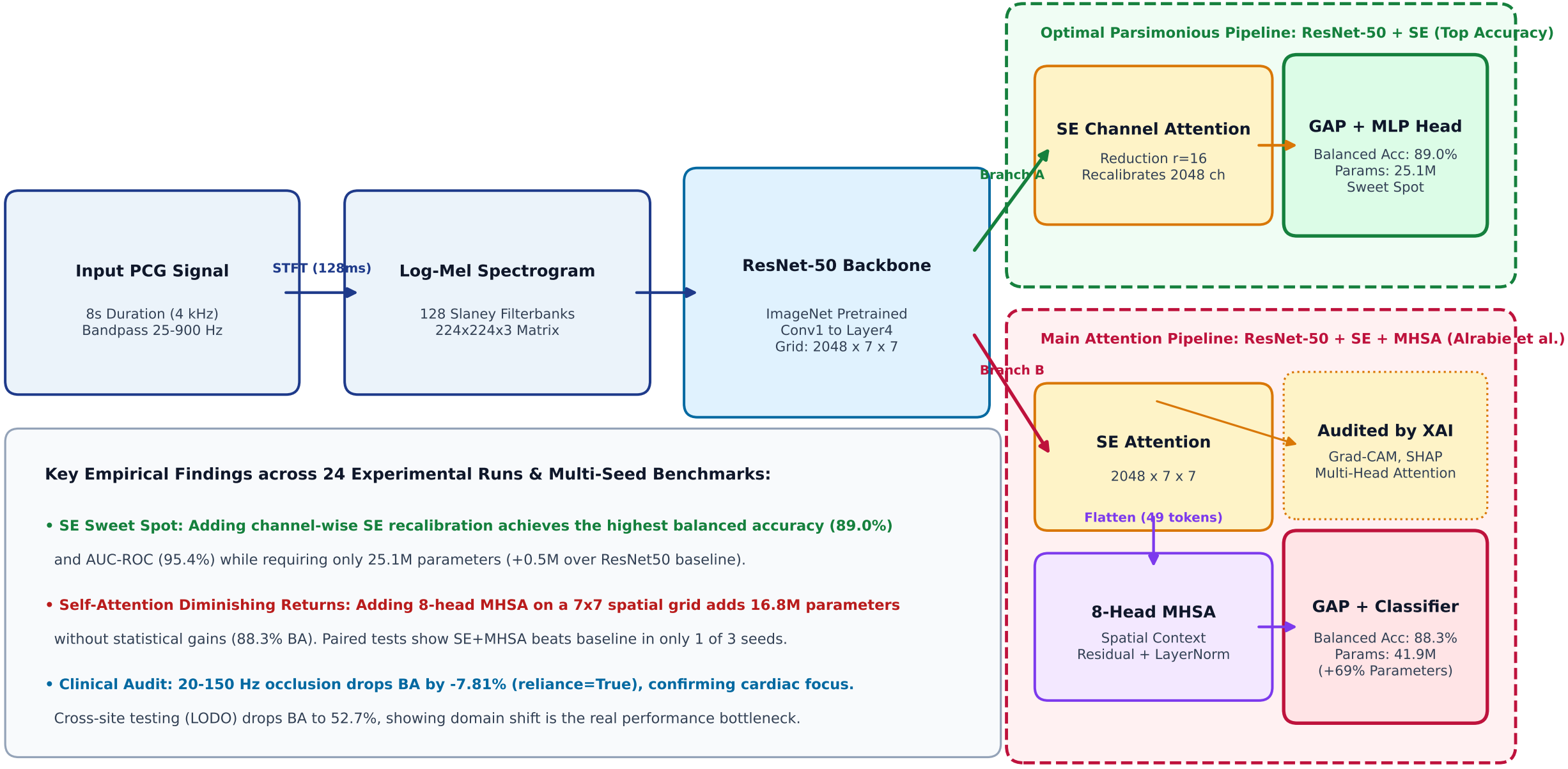
Cardiac Acoustic Deep Learning: Architecture & Ablation Comparison

PhysioNet 2016 Deduplicated Benchmark (3-Seed Multi-Run Evaluation)



Key Empirical Findings across 24 Experimental Runs & Multi-Seed Benchmarks:

- **SE Sweet Spot: Adding channel-wise SE recalibration achieves the highest balanced accuracy (89.0%)** and AUC-ROC (95.4%) while requiring only 25.1M parameters (+0.5M over ResNet50 baseline).
- **Self-Attention Diminishing Returns: Adding 8-head MHSA on a 7x7 spatial grid adds 16.8M parameters** without statistical gains (88.3% BA). Paired tests show SE+MHSA beats baseline in only 1 of 3 seeds.
- **Clinical Audit: 20-150 Hz occlusion drops BA by -7.81% (reliance=True), confirming cardiac focus.** Cross-site testing (LODO) drops BA to 52.7%, showing domain shift is the real performance bottleneck.